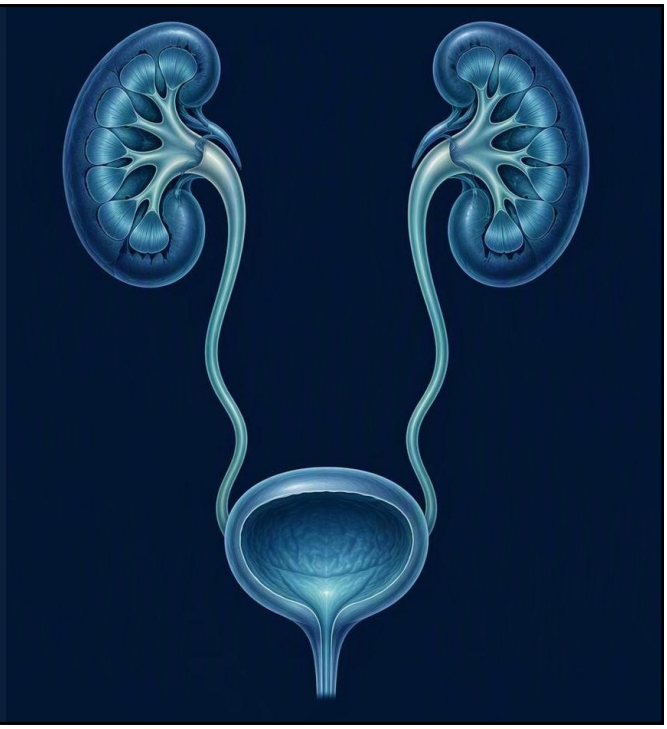
MEDICAL GENETICS • URINARY SYSTEM

Genetic Disorders of the Human Urinary Tract

From embryologic development to precision diagnosis



Assist. Prof. Dr. Evan Latef Khlef

2026

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LEARNING OBJECTIVES

02

By the end, you should be able to...

01 **Explain**
how ureteric bud–mesenchyme signaling builds the kidney and collecting system

02 **Recognize**
major CAKUT and inherited cystic

03 **Predict**
autosomal dominant, autosomal recessive, and X-linked recurrence patterns

04 **Select**
an appropriate genetic test and interpret a VUS cautiously

05 **Counsel**
patients about relatives, surveillance, and reproductive options

Course map: development → phenotype → inheritance → testing → counseling

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03

GENETICS OF THE KIDNEY AND URINARY TRACT

From Genetic Defects to Kidney Disease

How genetic variants can affect kidney development and function

1. Kidney formation
Biological process
Induction and tissue formation
Disorders
Renal agenesis; renal dysplasia
Associated genes
GREB1L, HNF1B and others

2. Kidney growth
Biological process
Branching and nephron formation
Disorders
Renal hypoplasia; renal dysplasia
Associated genes
Several possible genes; depends on the phenotype

3. Urine drainage
Biological process
Urinary tract connection
Disorders
Vesicoureteral reflux (VUR); congenital obstruction
Associated genes
Multiple genes may contribute

3

04

GENETICS OF THE KIDNEY AND URINARY TRACT

From Genetic Defects to Kidney Disease

How genetic variants can affect kidney development and function

4. Specialized cell function
Biological process
Glomerular structure and filtration barrier
Disorder
Alport syndrome
Associated genes
COL4A3, COL4A4, COL4A5

5. Tubular and ciliary function
Biological process
Tubular integrity and cell signaling
Disorder
Nephronophthisis (NPHP)
Associated genes
NPHP1 and other NPHP genes

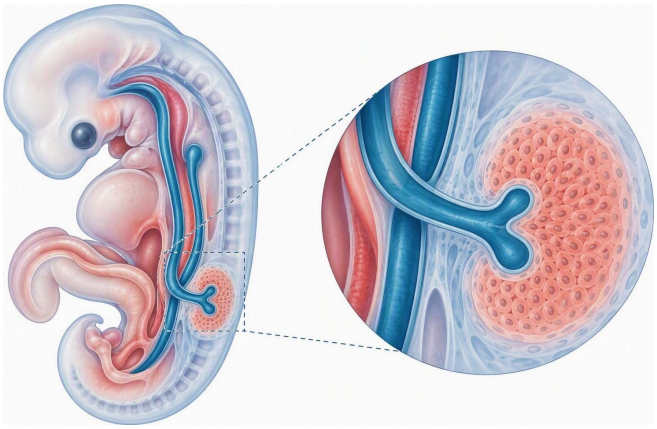
6. Cellular signaling
Biological process
Polycystin signaling and cyst formation
Disorder
Autosomal dominant polycystic kidney disease (ADPKD)
Associated genes
PKD1, PKD2

4

EMBRYOLOGY

05

Kidney development depends on reciprocal induction



4–5 weeks

Ureteric bud sprouts from mesonephric duct

5–10 weeks

Branching morphogenesis + nephron induction

~34–36 weeks

Nephrogenesis largely completes before birth

Two tissues, one organ

Collecting system ← ureteric bud
Nephrons ← metanephric mesenchyme

Sources: Kolvenbach et al., Nat Rev Nephrol 2023; Vivante & Hildebrandt 2021.

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DEVELOPMENTAL GENETICS

06

Five regulatory nodes organize urinary tract development

WT1	**Supports kidney development and differentiation**
GDNF → RET	**Promotes ureteric bud formation and outgrowth**
PAX2	**Supports kidney and urinary tract development**
HNF1B	**patterns tubules and multiple organs**
Branching signals	**balance growth and differentiation**

Genes are pathway markers—not one-gene/one-malformation labels.

Source: Kolvenbach et al., Nat Rev Nephrol 2023.

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CAKUT

07

CAKUT is a spectrum, not one diagnosis

12–20%

of CAKUT may have an identified monogenic cause

4–11%

may have a pathogenic copy-number variant

NUMBER / POSITION

agenesis • ectopia • horseshoe kidney

PARENCHYMA

hypoplasia • dysplasia • cystic dysplasia

COLLECTING SYSTEM

duplex system • hydronephrosis • UPJ obstruction

OUTFLOW / REFLUX

posterior urethral valves • VUR

Why the yield varies

Phenotyping depth • syndromic clues • family history • test platform

Source: Kolvenbach et al., Nat Rev Nephrol 2023. Yields are cohort- and method-dependent.

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RENAL AGENESIS

08

Renal agenesis ranges from incidental to life-limiting

UNILATERAL

Often asymptomatic

Compensatory hypertrophy of the solitary kidney

Look for

- contralateral CAKUT
- genital tract anomalies
- hypertension / albuminuria over time

Counsel: protect the solitary functioning kidney and screen relatives selectively.

BILATERAL

Severe fetal phenotype

Anuria → oligohydramnios → pulmonary hypoplasia

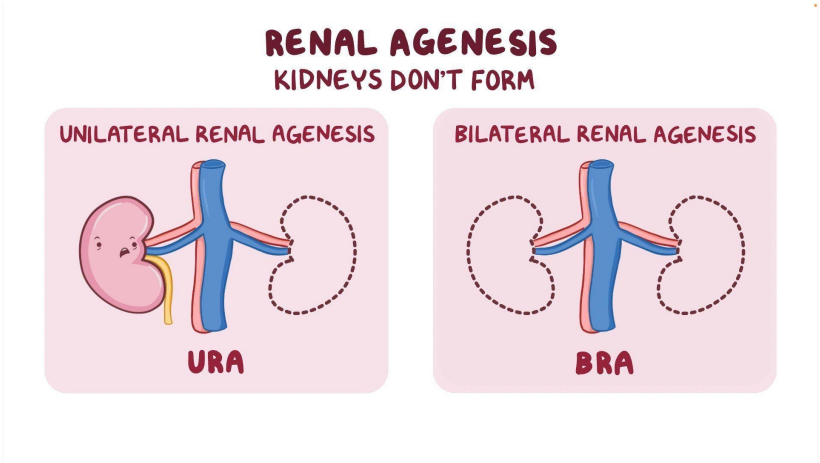
Prenatal clues

- absent kidneys / renal arteries
- non-visualized bladder
- reduced amniotic fluid

Counsel: confirm anatomy, discuss prognosis, recurrence, and genomic evaluation.

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09



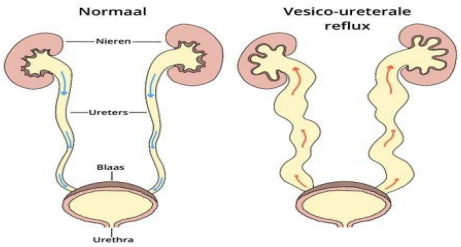
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VESICOURETERAL REFLUX

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Vesico-uerterate reflux (VUR) links anatomy, infection, and renal scarring

Short intramural ureteric tunnel → **Retrograde urine flow** → **Febrile UTI / pyelonephritis** → **Scarring risk in susceptible kidneys**



Management is risk-based

Age • febrile UTIs • grade • bladder–bowel dysfunction • renal scars • adherence

Genetics

Familial aggregation is common, but most isolated VUR is genetically complex.

Sources: AUA VUR Guideline; EAU Paediatric Urology Guideline; Ninoa et al., Curr Genomics 2016.

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CLINICAL CASE 1

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The newborn with oligohydramnios

TERM INFANT

Absent left kidney • small echogenic right kidney • no other obvious anomaly

1	2	3	4
Stabilize + phenotype	**Look beyond kidney**	**Choose testing**	**Interpret in family**
Urine output, creatinine trend, ultrasound, blood pressure	Genital exam, spine, heart, hearing, eyes; review prenatal history	Chromosomal microarray + phenotype-driven CAKUT panel / exome	Parental imaging and segregation can clarify recurrence

Best next genetic move — **Test the infant—not only the parents—and pair sequence analysis with copy-number detection.**

Teaching case; details are illustrative. Testing strategy: Kolvenbach et al., 2023.

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INHERITANCE

12

Inheritance changes recurrence risk

AUTOSOMAL DOMINANT	AUTOSOMAL RECESSIVE	X-LINKED
● × ○	◐ × ◐	■ × ○
Typical risk per pregnancy	Typical risk per pregnancy	Typical risk per pregnancy
1 in 2	**1 in 4**	**sex-dependent**
ADPKD • HNF1B • PAX2 • many WT1 variants	**ARPKD • nephronophthisis**	**COL4A5-related Alport syndrome**
Variable expression; de novo variants are common in some disorders	Parents are usually unaffected carriers; consanguinity may increase prior probability	Transmission and severity differ by sex and variant; avoid the term “benign carrier”

Always modify the textbook risk for de novo variation, reduced penetrance, parental mosaicism, and the actual molecular result.

Symbols are schematic: filled = affected; half-filled = carrier; open = unaffected.

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GENETIC TESTING

13

Genetic testing works best as a phenotype-first ladder

1 • PHENOTYPE	Imaging, urine, hearing, eyes, liver, glucose, magnesium, pedigree
2 • COPY NUMBER	Chromosomal microarray or validated CNV calling when CAKUT is syndromic
3 • SEQUENCE	Focused panel when phenotype is specific; exome/genome when heterogeneous
4 • INTERPRET	ACMG classification + phenotype fit + segregation; a VUS is not a diagnosis
5 • ACT	Translate the result into surveillance, treatment, family testing, and counseling

NEGATIVE ≠ NON-GENETIC

Reanalysis matters as genes, classifications, and phenotypes evolve.

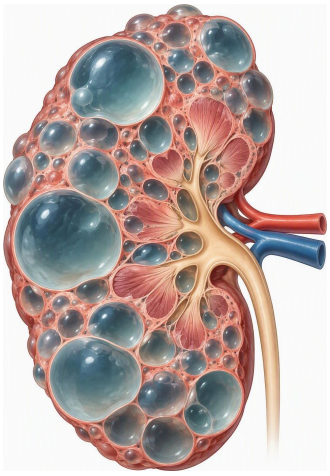
Use a laboratory method that detects the variant classes relevant to the suspected disorder.

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ADPKD

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Autosomal Dominant Polycystic Kidney Disorder (ADPKD) is systemic and age-dependent



PKD1 • PKD2

Autosomal dominant; cyst burden increases with age

KIDNEY	Bilateral cysts, hypertension, pain, hematuria, stones, progressive CKD
BEYOND KIDNEY	Liver cysts; selected patients need intracranial aneurysm risk assessment
DIAGNOSIS	Age-specific imaging criteria; genetic testing when imaging/family history is equivocal or for reproductive/donor decisions
MANAGEMENT	Blood-pressure control, CKD care, lifestyle, and disease-modifying therapy for appropriate rapidly progressive disease

Source: KDIGO 2025 ADPKD Guideline. Testing and therapy should follow current specialist guidance.

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CLINICAL CASE 2

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A 28-year-old with hypertension and family history

BP 152/94

eGFR normal • bilateral renal cysts • father reached kidney failure at 54

1	2	3	4	5
CONFIRM	STRATIFY	TEST?	COUNSEL	ACT
Review imaging pattern and family history	Kidney size, eGFR trend, BP and complications	Useful if atypical, family variant unknown, or planning	50% transmission risk; respect right not to know	Treat BP; assess progression risk

Teaching case; principles align with KDIGO 2025 ADPKD guidance.

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ARPKD

16

Autosomal Recessive (ARPKD) collecting-duct disease with hepatic fibrosis

KIDNEY

Enlarged, echogenic kidneys

Dilated collecting ducts • neonatal respiratory compromise • childhood hypertension • progressive CKD

Prenatal ultrasound may detect enlarged echogenic kidneys and reduced amniotic fluid.

LIVER

Congenital hepatic fibrosis

Portal hypertension • splenomegaly • varices • recurrent cholangitis in some patients

Liver disease may dominate later even when renal function is relatively preserved.

PKHD1 ◐ × ◐ → **25% affected per pregnancy** Autosomal recessive

Source: NCBI Bookshelf ARPKD review; molecular confirmation supports prognosis and family planning.

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PATTERN RECOGNITION

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ADPKD and ARPKD separate by pattern—not cysts alone

FEATURE	ADPKD	ARPKD
Inheritance	Autosomal dominant	Autosomal recessive
Typical timing	Adolescence/adulthood; age-dependent	Prenatal, neonatal, or childhood
Kidney pattern	Bilateral variable-sized cysts; enlarged kidneys	Enlarged echogenic kidneys; collecting-duct ectasia
Liver pattern	Cysts / polycystic liver disease	Congenital hepatic fibrosis / portal hypertension
Genes / test	PKD1, PKD2; consider phenocopies	PKHD1; broad cystic panel if atypical

Sources: KDIGO 2025 ADPKD Guideline; NCBI Bookshelf ARPKD review.

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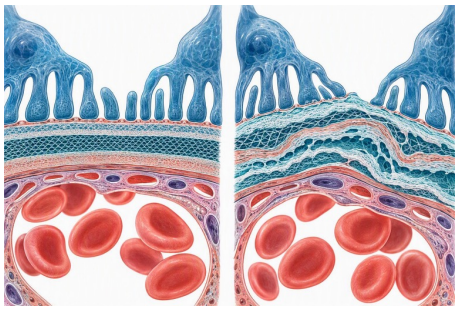
ALPORT SYNDROME

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Alport syndrome is a collagen IV basement-membrane disorder

COL4A3 • COL4A4 • COL4A5

Abnormal α3α4α5 type IV collagen



KIDNEY — persistent hematuria → proteinuria → CKD

EAR — sensorineural hearing loss

EYE — lenticonus and characteristic retinal changes in some patients

Inheritance — X-linked is common • autosomal recessive and autosomal dominant forms also occur

Source: Torra et al., Alport syndrome guideline 2024/2025.

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CLINICAL CASE 3

19

Persistent hematuria is the diagnostic clue

14-year-old boy

Microscopic hematuria since age 6 • new albuminuria • normal complement

FAMILY			
Maternal grandfather: kidney failure at 42	Mother: hematuria	Patient: high-frequency hearing loss	

SUSPECT	TEST	ACT	CASCADE
X-linked Alport syndrome	COL4A3/4/5 sequence + deletion/duplication analysis	Nephrology care, hearing/eye assessment, early kidney-protective therapy	Offer targeted testing and surveillance to at-risk relatives

Teaching case. Source: Torra et al., Alport syndrome guideline 2024/2025.

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